TravelTide Rewards: Detailed Report

Prepared for Elena Tarrant, Head of Marketing, and the marketing team | Jacklynn Tong, Data Analyst | November 2026

1. Background and objectives

TravelTide is launching a customer rewards program designed to drive long-term retention, increase booking frequency, and boost customer lifetime value. Your core premise is that personalized marketing performs best when an invitation opens with the single perk a customer values most. From Marketing's experience, five perks should attract our customers: a free hotel meal, a free checked bag, no cancellation fees, exclusive discounts and one free hotel night with a flight.

To turn this vision into an actionable strategy, this analysis addresses two primary business questions:

1. **Perk validation:** Does our customers' booking behavior show that they would value these perks?
2. **Customer segmentation:** How can we assign every customer in the cohort to a distinct segment and map them to their likely favorite perk?

Using our agreed-upon cohort of 5,998 highly engaged customers (49,211 sessions and 15,489 trips they didn't cancel), I evaluated your proposals against what chance alone would produce. Three of your five proposed perks are backed by genuine customer habits, two look like random chance and should be tested before scaling, and the data reveals one brand-new perk opportunity for long-distance travelers. The main deliverable is `user_perks.csv`: one perk for each of the 5,998 customers, plus a pilot idea per segment where the data pointed to something we hadn't planned.

2. Data and method

Cohort. We focused on customers with more than seven sessions since January 4, 2023, narrowing the database from 1,020,926 users down to 979,801 active users and then to 5,998 engaged customers, with 49,211 sessions up to July 28, 2023. One SQL query (`sql/session_and_user_base.sql`) builds the session table and the user table, and notebook 01 can run it directly against the database.

Data preparation. A few decisions here change the results, so they are worth spelling out:

- **Canceled trips:** a canceled trip shows up again in a second session. I left all 610 canceled trips out of the travel metrics, so each of the 15,489 kept trips counts once (cancellation rate 3.8 %).
- **Hotel nights:** the raw nights field counts full 24-hour periods and even has negative values, so I recalculated nights from the check-in and check-out dates. The 2.9 % of hotel bookings with invalid dates are flagged and their spend is left as unknown.
- **Discounts:** the session flags also record offers on the part of a trip that wasn't booked. Counting a discount only on the booked part corrected 549 trips and lowered the share of discounted trips from 26.3 % to 22.7 %.
- **Birth year 2006:** 233 users have this birth year, compared with about 51 for each earlier year, so it looks like a placeholder. These users are flagged; age plays no part in the perks.
- **Outliers:** long-haul flights and large parties are real trips, so I kept them. They help define segments.

Features. I built 35 user-level features from kept trips, such as trips, party size, nights, hotel-only share, flight distance, discount share and repeat visits to the same city.

Segmentation in four steps:

1. **Rules in priority order.** A customer can match more than one rule, so the rules are checked in a fixed order: the most specific habits first (no kept trip, families, couples and friends, hotel-only stays, favorite cities) and broad behaviors found in every group (discounts, long flights, packages) last, so they don't absorb a more telling habit. The first match decides, so every customer lands in exactly one of nine segments.
2. **Habit or luck?** For each rule, I shuffled all trips between customers 50 times and counted again. A rule counts as a habit when the real data has more matching customers than 95 % of the shuffles.
3. **Perk scores.** Each perk gets a standardized score. Customers who match no rule (the All-rounders) receive the perk with their highest score.
4. **Machine-learning cross-check.** I ran K-Means (k = 8), DBSCAN and PCA on nine standardized behavior features, without telling them about the rules, to test whether unsupervised clustering would surface similar groups (section 4).

3. Question 1: Does the data support the five proposed perks?

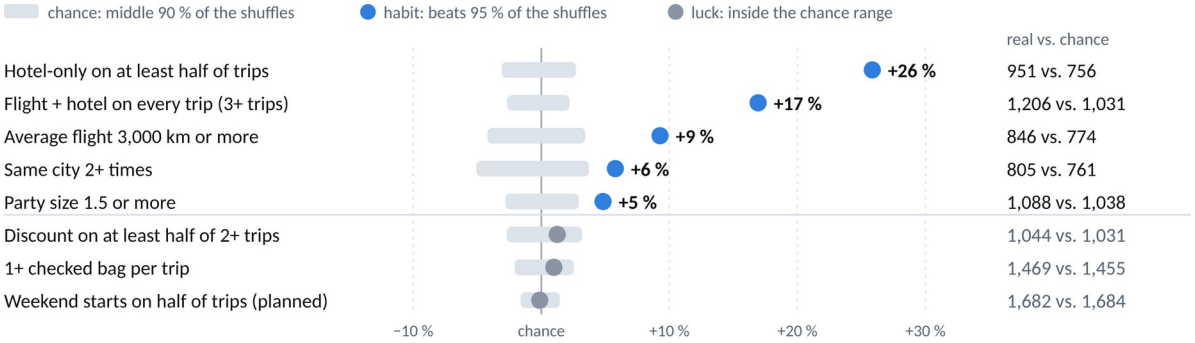
Partly. Three of the five perks rest on a real habit, two don't, and the data points to one new perk.

Proposed perk	Behavior tested	Customers: real vs. chance	Verdict
Free hotel meal	at least half of all trips hotel-only	951 vs. 756 (+26 %)	Supported, launch
1 night free hotel with flight	every trip flight + hotel, at least 3 trips	1,206 vs. 1,031 (+17 %)	Supported, launch
No cancellation fees	one booking, then canceled	99 vs. 36 (about 3×)	Supported for new customers, launch

Proposed perk	Behavior tested	Customers: real vs. chance	Verdict
Exclusive discounts	discount on at least half of 2+ trips	1,044 vs. 1,031 (+1 %)	Luck, A/B test first
Free checked bag	1+ checked bag per trip on average	1,469 vs. 1,455 (+1 %)	Luck, test only
New: Extra-legroom seat	average flight 3,000 km or more	846 vs. 774 (+9 %)	Supported, launch

Five segment rules beat chance; discounts, checked bags and weekend trips don't

Customers who match each rule in the real data, compared with 50 random shuffles of all 15,489 kept trips (notebook 03)



How to read the chart: each dot compares the customers who match a rule with the average of the 50 shuffles, and the grey bar is what chance alone produces (the middle 90 % of the shuffles). A dot to the right of the bar beats 95 % of the shuffles, so that behavior is a habit; a dot inside it could just be luck.

Key takeaways:

- **The power of packages and stays:** customers who mostly book hotel-only trips and customers who always book flight + hotel clearly beat chance (+26 % and +17 %), which supports the free hotel meal and the free hotel night. First-time bookers also cancel about three times as often as chance predicts, which makes no cancellation fees a strong hook for converting new customers.
- **The discount paradox:** discount use looks like chance however it is measured (frequency, depth and customer response). Sessions with a flight discount even ended in a flight booking less often (23.8 % vs. 29.1 %), and a customer's booking rate shows no correlation with how often they use discounts ($r = -0.02$). The data suggests that discounts don't build booking habits, hence the recommendation to test them against a control group first. One caveat: we only see past offers made by TravelTide, not how customers would respond to a guaranteed perk.
- **Perk independence:** the perk scores are barely correlated ($|r| \leq 0.12$). The single exception, the free hotel meal vs. the free hotel night ($r = -0.52$), is a mirror image rather than a duplicate: one fits hotel-only travelers, the other package bookers, so we need both.
- **The long-haul discovery and unplanned patterns:** distance emerged as a real habit while checked bags did not, which gave rise to the extra-legroom seat. Our long-haul customers mostly fly coast to coast (97 % of flights over 3,000 km stay within North America) but don't check more bags than average. Also, 458 customers never book, about twice as many as chance predicts (236); these Dreamers get no cancellation fees, plus a 48-hour fare lock as a pilot. Traveling with companions is a habit too (party of 2 or more: 413 vs. 364, +13 %), whereas weekend trips proved to be pure chance (1,682 vs. 1,684), so the planned Weekend Escapers segment was dropped.

4. Question 2: One perk for every customer

All 5,998 customers have a perk: 4,561 (76 %) through a segment rule and 1,437 All-rounders (24 %) through their perk scores.

Segment	Rule (first match wins)	Customers	Perk	Pilot perk
Dreamers	no kept trip yet	556	No cancellation fees	48-hour fare lock
Family Crew	children, party size 1.5+	417	Free hotel meal	Family fare
Couples & Friends	no children, party size 1.5+	829	No cancellation fees	Bring-a-friend
Hotel Stayers	at least half of trips hotel-only	616	Free hotel meal	Welcome dinner on night 1
Favorite-City Returners	same city 2+ times	604	1 night free hotel with flight	City Stamp Card
Deal Hunters	discount on half of 2+ trips	517	Exclusive discounts	Price-drop guarantee
Long-haul Travelers	average flight 3,000 km+	377	Extra-legroom seat	Free checked bag
Package Travelers	every trip flight + hotel, 3+ trips	645	1 night free hotel with flight	Free airport transfer
All-rounders	no rule matched	1,437	highest perk score	–

Among the All-rounders, the highest scores lead to a free checked bag (506), a free hotel night (306), a free hotel meal (238), exclusive discounts (206) and no cancellation fees (181). As a check, the highest score matches the rule-based perk for 57 % of segment members.

Operational insights on segments:

The segments reflect habits, but matching a perk to a segment still takes some judgment. **Couples & Friends** get no cancellation fees to encourage more risk-free booking for group travelers, since a trip with several people is harder to commit to. Their own cancellation rate (3.3 %) is actually below the cohort average (3.8 %), so this perk is about winning the booking and reducing hesitation when a group coordinates, rather than about refunds.

Machine-learning cross-check. Customer behavior turns out to be a continuum with a few distinct groups at the edges, which is why transparent rules work better than clusters for assigning perks:

- **K-Means** (k = 8, silhouette 0.18) found 4 of the 7 booker segments on its own, most clearly Favorite-City Returners and Hotel Stayers. It groups the rest mainly by how often they travel.
- **DBSCAN** flags 97 % of the 115 most extreme customers as outliers, but it puts two thirds of all bookers into one cluster, so it can't work as a segmentation on its own.
- **PCA** needs 7 of the 9 components to explain 80 % of the variance, so there is no shortcut: **each feature adds information**.
- **What clustering can't tell us:** it shows who is similar, but not whether that is a habit. K-Means even forms a discount cluster (53 % Deal Hunters), although discount use is luck. That's why a perk launches only if the behavior behind it passed the shuffle test; the others are tested first.

Who our customers are. The cohort is 88 % women (44 % among all users), a side effect of the more-than-seven-sessions rule. Neither gender nor age drives the perks (age vs. behavior: $|r| \leq 0.11$), and only the Dreamers stand out as younger (median age 28).

5. Recommendations and how to measure success

Action plan for the marketing team:

1. **Launch now (4,769 customers, 80 %):** roll out the four habit-backed perks (no cancellation fees, the free hotel night with a flight, the free hotel meal and the extra-legroom seat). Every invitation leads with the customer's own perk from `user_perks.csv`.
2. **Test first (1,229 customers, 20 %):** start exclusive discounts and the free checked bag in a test phase against a control group, so they prove they pay off before we scale them.
3. **Pilot one new idea per segment (section 4):** start with the 48-hour fare lock for the Dreamers, which takes the risk out of a first booking.
4. **Re-run the segmentation every quarter** with new data: the SQL query and the notebooks rebuild everything from the database.

Test design. Group sizes and test length are proposals to adjust to TravelTide's traffic. For launched perks, a random 20 % of each perk group gets no perk for three months; tested perks get a 50/50 split. After three months, we keep what beats the control group and switch the rest to the pilot perk.

To evaluate the program over those three months, we track these KPIs against today's baselines:

KPI	Today (baseline)	Definition
Rewards sign-ups	new	main KPI: share of invited customers who sign up
Booking rate	33.1 %	shopping sessions that end in a booking
Trips per customer	2.6	15,489 kept trips ÷ 5,998 customers
Cancellation rate	3.8 %	610 of 16,099 booked trips
Avg spend per trip	$1,188	flight + hotel, after discounts

6. Assumptions and limitations

- **Six and a half months of data** (January 4 to July 28, 2023). Customers book about a week ahead (median 8 days), so trips from August on are mostly missing (224 trips, 1.4 %), and seasons can't be compared.
- **Engaged customers only.** The cohort is 5,998 of about 1 million users, so the results describe TravelTide's most active customers rather than the whole user base.
- **Behavior stands in for preference.** The data shows what customers did, not which perk they would pick. I assume that a repeated behavior points to the perk a customer values most, and the A/B tests will check this assumption.
- **Few trips per customer.** With 2.6 trips on average, single rules can be noisy, which is why every rule had to beat chance first.
- **Data quality.** 421 trips with invalid hotel dates have unknown spend, and the 233 placeholder birth dates are flagged. Neither changes a perk.

Files: notebooks 01 to 03, the SQL files and `user_perks.csv` are on GitHub (github.com/jtong28/TravelTide), and the interactive story is on Tableau Public (public.tableau.com/views/TravelTideRewards_JT/TravelTide).